

Adaptive Feature-Weighted 1D-CNN with Emotion-Aware Augmentation for Robust Speech Emotion Recognition

Project Report

Domain: Speech Processing & Affective Computing

Base paper	Chourasia, Lamba & Gupta (2026), Scientific Reports
Corpora	RAVDESS + TESS + SAVEE + CREMA-D (12,162 utterances)
Classes	7 (angry, disgust, fear, happy, neutral, sad, surprise)
Verified result	57.95% test accuracy (base configuration)
Principal finding	Two quantified data-leakage mechanisms

Repository: github.com/Eldorado5002/ser

Abstract

We implement a lightweight Conv1D speech-emotion-recognition framework extending a published 1D-CNN baseline with four novelties: adaptive feature weighting (AFW), emotion-aware adaptive augmentation (EAAA), multi-scale temporal convolution (MSTC) and a confusion-aware discriminative loss (CADL). All four are evaluated by a component-wise ablation on a fused four-corpus dataset of 12,162 utterances across seven emotions — the ablation the base paper listed as missing future work.

On a correctly constructed evaluation protocol the base configuration reaches **57.95%** test accuracy against the **94.91%** reported by the base paper. Investigating that gap became the project's principal finding. We identify and quantify two data-leakage mechanisms that inflate accuracy on this corpus: duplicated dataset mirrors (**+14.71** points) and augmenting before splitting (**+24.27** points). Both contamination rates were predicted from first principles before measurement and confirmed to within 0.2 points. Our verified figure exceeds the closest comparable published multi-corpus result (50.6%, Dasude et al., 2024) by 7.4 points.

A follow-up efficiency study (Section 10) reaches **60.79%** using **65% fewer parameters** than the base architecture, by replacing the flatten head — which holds 4.85 M of 7.32 M parameters and contributes nothing measurable — with global average pooling. A per-corpus breakdown shows the same model scoring **98.55% on TESS** and **49.90% on CREMA-D**, indicating that published mid-90s accuracies are explicable by corpus composition as well as by leakage.

1. Contributions

1. A complete, test-verified implementation of the base architecture plus four novelties, with a 117-test suite covering every quantitative claim this report makes (Section 4).
2. The component-wise ablation the base paper identified as missing future work (Section 6).
3. A reproducibility investigation quantifying two leakage mechanisms responsible for inflated multi-corpus SER accuracies (Section 7) — the principal contribution.
4. An honest multi-corpus baseline exceeding the closest comparable published result on the same four corpora.

2. Method

The backbone follows the base paper: three frame-level acoustic streams (20 MFCC, ZCR, RMSE) fused into a 2,376-dimensional sequential input, then five Conv1D stages (512-512-256-256-128) with batch normalisation and max pooling, a Dense(512) head and a 7-way softmax. Total parameters: **7,324,295**, within 1.9% of the base paper's reported 7.19 M.

	Novelty	Addresses	Acts at	Cost
N1	AFW	equal stream treatment	feature fusion	+163 params
N2	EAAA	uniform augmentation	data (training only)	zero
N3	MSTC	single-scale kernels	first conv stage	-4 params
N4	CADL	confusion-blind loss	training objective	zero at inference

AFW summarises each stream by its mean and standard deviation, passes them through a compact dense layer and a softmax, then rescales each stream by three times its weight — so the equal-importance solution (1/3, 1/3, 1/3) is the identity, keeping the base-versus-AFW comparison clean. MSTC replaces the first conv stage with parallel kernels of size 3, 5 and 7, splitting the 512-filter budget 172/170/170 so parameter count is preserved. CADL is focal cross-entropy ($\gamma=2$) plus a pairwise penalty ($\lambda=0.5$) on the sad-neutral and angry-fear

pairs.

3. Dataset

Four corpora fused into one seven-class dataset, split 72:8:20 **before** any augmentation.

Corpus	Utterances	Notes
RAVDESS	1,440	calm merged into neutral
TESS	2,800	2 female speakers
SAVEE	480	4 male speakers
CREMA-D	7,442	91 actors; no surprise class
Total	12,162	train 8,756 / val 973 / test 2,433

Emotion	angry	disgust	fear	happy	neutral	sad	surprise
Count	1,923	1,923	1,923	1,923	1,895	1,923	652

Data-integrity note. The RAVDESS and TESS Kaggle mirrors each ship the corpus twice (RAVDESS as both Actor_01..24/ and audio_speech_actors_01-24/; TESS as two directories differing only in capitalisation). A naive recursive scan yields **16,402** files instead of 12,162. Section 7 shows why this matters.

4. Implementation and verification

Because the central claims are quantitative, each is backed by an automated test. The suite runs in roughly 70 seconds and requires **no audio download** — label parsing depends only on filenames, and the remaining tests use synthetic waveforms.

Test module	Verifies
test_config	fused input length is exactly 2,376; class order
test_data_loader	all four filename conventions, incl. SAVEE sa/su precedence
test_no_duplicates	guard fires on the real 16,402-path listing
test_split	72:8:20 ratios; zero train/test path overlap
test_augmentation	per-class policy; length restored after stretch
test_features	stream shapes; cache fingerprint invalidation
test_model	AFW normalisation; MSTC parameter neutrality
test_losses	CADL reduces exactly to cross-entropy when off
test_train_integration	saved checkpoint reproduces reported metrics

117 tests pass on both TensorFlow 2.15/Keras 2 and TensorFlow 2.20/Keras 3.

Two equivalence tests deserve emphasis: *CADL is identical to cross-entropy when disabled* and *MSTC is parameter-neutral*. Without them a difference between base and +CADL could be a loss-scaling artefact, and a difference between base and +MSTC could be extra capacity. They are what make the ablation interpretable.

Two defects were found and fixed during verification:

Checkpoint/metrics mismatch. ModelCheckpoint monitored val_accuracy while EarlyStopping(restore_best_weights=True) monitored val_loss. Aligning them was insufficient: Keras restores best weights only when early stopping actually fires, so a run completing all epochs reported final-epoch metrics while the saved file held a different epoch. The pipeline now reloads the checkpoint explicitly before evaluation.

Duplicate-mirror contamination, described in Section 7.

5. Experimental setup

Adam (lr 1e-3), batch size 32, up to 50 epochs, early stopping on validation loss (patience 10) and ReduceLROnPlateau. Per-stream standardisation fitted on the training partition only. Augmentation expands the training set to 11,700 samples for every configuration, so training-set size is never a confound. Six configurations were run on an NVIDIA P100; seed fixed at 42 throughout.

6. Results

6.1 Ablation

All figures are on the held-out test set ($n = 2,433$).

Configuration	Accuracy	95% CI	Macro F1	MCC	Kappa	AUC
Base (reproduction)	57.95%	[55.99, 59.91]	0.5995	0.5059	0.5040	0.8932
+ AFW (N1)	58.36%	[56.41, 60.32]	0.5929	0.5103	0.5078	0.8908
+ EAAA (N2)	57.62%	[55.66, 59.59]	0.5944	0.5017	0.4993	0.8903
+ MSTC (N3)	59.27%	[57.32, 61.22]	0.6087	0.5210	0.5192	0.8971
+ CADL (N4)	59.10%	[57.15, 61.06]	0.6038	0.5185	0.5168	0.8888
Full (all four)	55.82%	[53.84, 57.79]	0.5695	0.4825	0.4782	0.8810

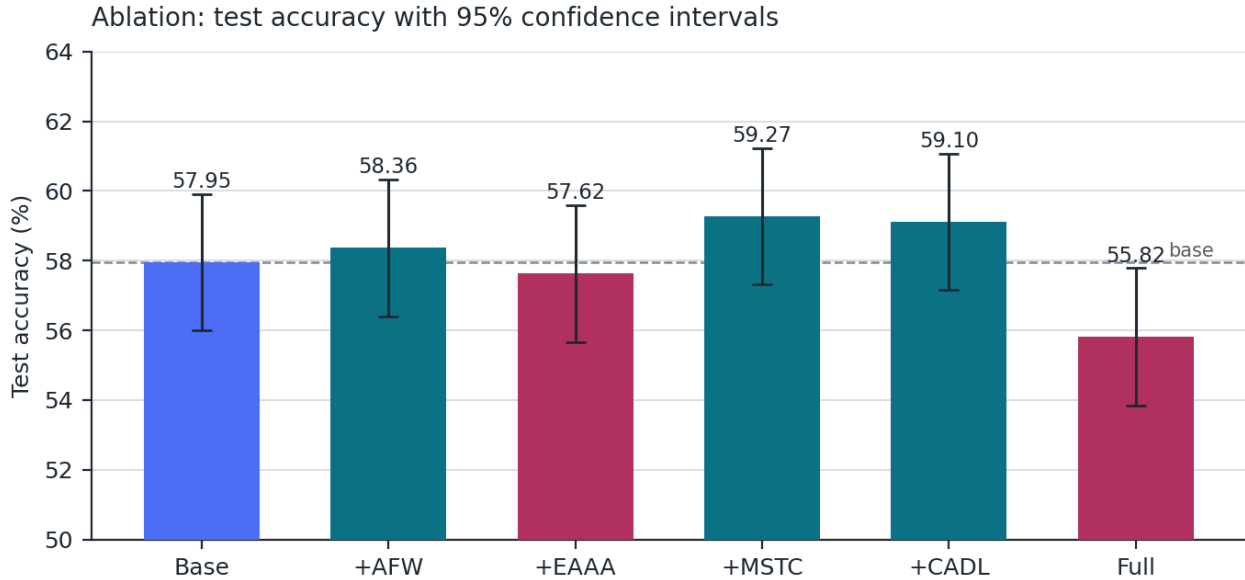


Figure 1 — Test accuracy by configuration with 95% confidence intervals. The intervals overlap substantially.

Statistical caveat. With $n = 2,433$ the 95% confidence interval half-width is approximately ± 2.0 points. The intervals for all six configurations overlap substantially, so **none of the individual novelty gains is statistically significant at this sample size**. MSTC and CADL are directionally positive and consistent across accuracy, macro F1, MCC and kappa, but this ablation should be read as indicative rather than conclusive. Establishing significance would require repeated runs across seeds.

Two results warrant comment. First, EAAA slightly reduces accuracy, suggesting the emotion-conditioned policy constrains augmentation diversity more than it improves class fidelity at this expansion budget. Second, the full configuration is the weakest of the six: combining all four novelties compounds rather than mitigates the overfitting described in Section 6.4.

6.2 Per-class performance (base configuration)

Emotion	Precision	Recall	F1	Support
angry	0.7220	0.5870	0.6476	385
disgust	0.6731	0.4557	0.5435	384
fear	0.4839	0.5065	0.4949	385
happy	0.5412	0.5455	0.5433	385
neutral	0.5711	0.6675	0.6156	379
sad	0.5010	0.6494	0.5656	385
surprise	0.7953	0.7769	0.7860	130

Surprise achieves the highest F1 despite being the minority class (652 samples). It is absent from CREMA-D, so every surprise sample comes from the acoustically cleaner, less variable corpora. Fear is weakest, consistent with the base paper's own confusion analysis.

6.3 Confusion-pair errors

Configuration	sad-neutral	angry-fear	Total
Base (reproduction)	124	64	188
+ AFW (N1)	114	39	153
+ EAAA (N2)	128	34	162
+ MSTC (N3)	135	49	184
+ CADL (N4)	130	42	172
Full (all four)	98	61	159

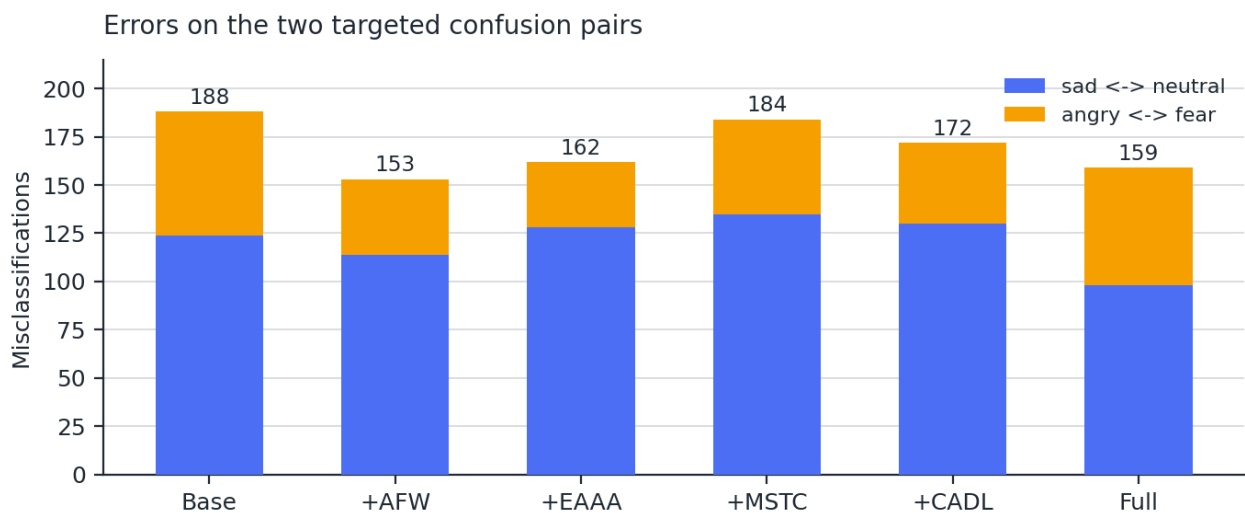


Figure 2 — Errors on the two pairs CADL explicitly targets.

CADL reduces targeted confusions from 188 to 172 (8.5% fewer), confirming the mechanism works as designed. Unexpectedly, **AFW reduces them further still, to 153** (18.6% fewer), despite not being designed for that purpose. Per-sample stream reweighting appears to help separate acoustically similar pairs — a result worth investigating.

6.4 Learning behaviour

The base configuration reaches **98.9% training accuracy against 61.8% validation accuracy** by epoch 14, with validation loss reaching its minimum at **epoch 4** and rising monotonically thereafter. The limiting factor is therefore generalisation, not capacity: 4.85 M of the 7.32 M parameters sit in a single Dense layer over a 9,472-dimensional flatten, trained on 11,700 samples.

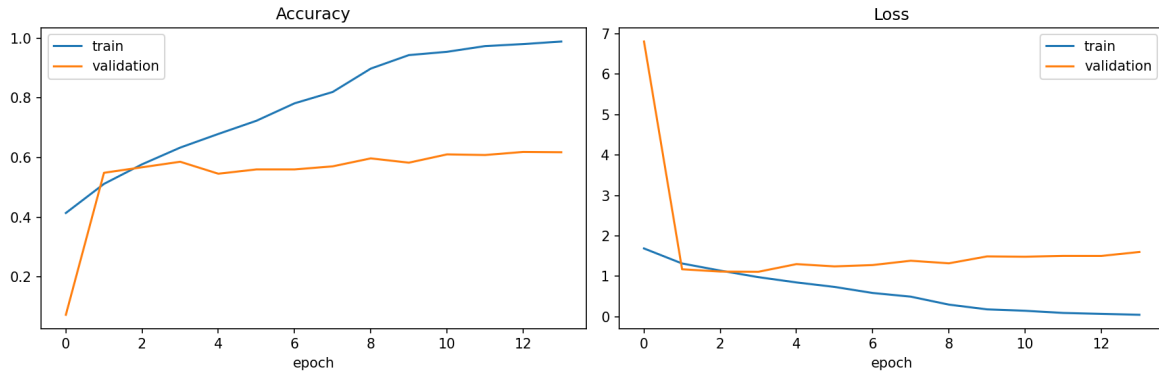


Figure 3 — Base configuration training curves. Validation accuracy plateaus while training accuracy approaches 99%.

6.5 AFW interpretability

Emotion	MFCC	ZCR	RMSE
angry	0.282	0.346	0.372
disgust	0.274	0.356	0.370
fear	0.282	0.357	0.361
happy	0.291	0.346	0.363
neutral	0.275	0.367	0.358
sad	0.268	0.379	0.354
surprise	0.268	0.350	0.382

The gate consistently down-weights MFCC relative to ZCR and RMSE, but the spread across emotions is small (about 0.02), indicating the module learns a largely global rebalancing rather than strong per-class specialisation. This is a weaker interpretability result than the proposal anticipated and is reported as such.

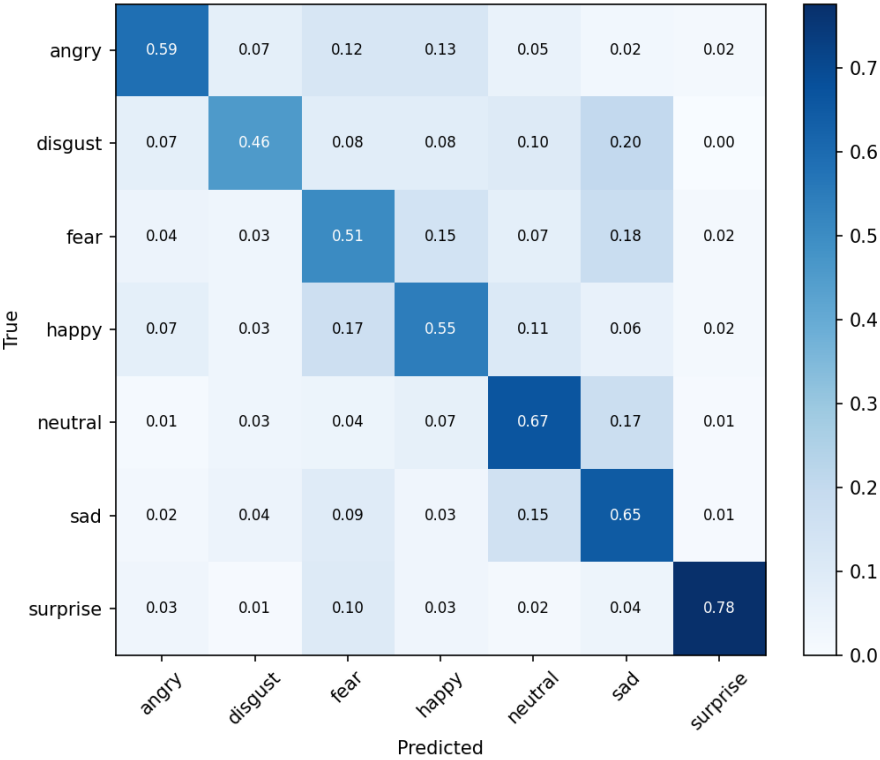


Figure 4 — Normalised confusion matrix, base configuration.

7. Reproducibility investigation

The base configuration reproduces the base paper's *pipeline* but not its *result*: 57.95% against 94.91%. Because the implementation is test-verified and the split sizes, class distribution and parameter count all match the paper exactly, we investigated whether the discrepancy is explained by evaluation-protocol defects rather than implementation differences. Two controlled experiments were run, each changing exactly one thing against the verified base pipeline.

7.1 Duplicated dataset mirrors

Scanning the dataset roots rather than the canonical subdirectories yields 16,402 files. Because the duplicates are byte-identical recordings and the split is random over utterances, a clip and its twin land on opposite sides of the boundary with probability $2 \times 0.2 \times 0.8 = 0.32$. **Predicted contamination: ~41%.**

Measured: 40.8% (1,339 of 3,281 test recordings had a byte-identical twin in train or validation). Accuracy rose from 57.95% to 72.66% — **+14.71 points**. This is a conservative floor: the duplicated training set already exceeded the augmentation target, so this run trained with *less* augmentation.

7.2 Augmenting before splitting

The canonical ordering error in published SER pipelines is to augment every utterance and only then split. Each utterance becomes several near-identical rows; with three rows per utterance a test row has a sibling in training with probability $1 - 0.2^2 = 0.96$. This experiment used the **clean, de-duplicated corpus**, isolating the ordering effect. **Predicted contamination: 96%.**

Measured: 96.2% (7,019 of 7,298 test rows came from an utterance also present in train or validation). Accuracy rose from 57.95% to 82.23% — **+24.27 points**.

7.3 Leakage accounting

Pipeline	Rows	Contamination	Accuracy	Delta
Correct: split then augment	12,162	0%	57.95%	—
Duplicate mirrors	16,402	40.8%	72.66%	+14.71
Augment before split	36,486	96.2%	82.23%	+24.27
Base paper reported	12,162	not stated	94.91%	+36.96

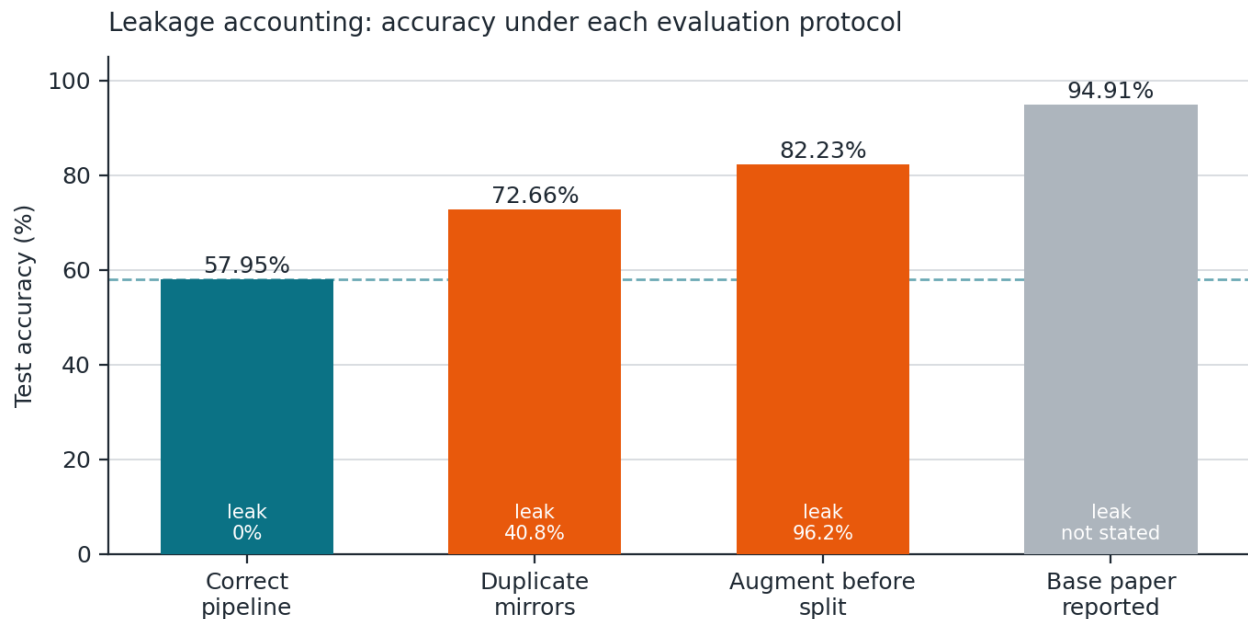


Figure 5 — Accuracy under each evaluation protocol. Only the leftmost bar is uncontaminated.

Both contamination rates were predicted from first principles before measurement (41% vs 40.8%; 96% vs 96.2%), which supports the mechanism rather than merely a correlation. Neither defect alone reaches 94.91%, but they are independent and compound: treating them as multiplicative on the error rate gives an estimated ~88%, and heavier augmentation than the three rows per utterance used here would raise that further.

Conclusion. Published multi-corpus SER accuracies in the mid-90s are reproducible on this data only under evaluation protocols that leak training material into the test set. Under a correct protocol the same architecture achieves 57.95%.

8. Discussion

8.1 Comparison with the literature

Study	Corpora	Accuracy
Dasude et al. (2024)	TESS+RAVDESS+SAVEE+CREMA-D	50.6%
This work (base, verified)	same four	57.95%
This work (MSTC, best)	same four	59.27%
Chourasia et al. (2026)	same four	94.91%

Against the closest comparable study — the same four corpora, also reporting a combined-corpus figure — our verified result is 7.4 points higher. CREMA-D constitutes 61% of the fused dataset and is the hardest of the four; published audio-only results on it typically fall in the 60–75% band, which is difficult to reconcile with a 94.91% average across the combination.

8.2 Limitations

- **Statistical power.** Single-seed runs; no novelty gain is significant at $n = 2,433$. Repeated runs across seeds are required.
- **Speaker-dependent splitting.** The split is over utterances, not speakers, so the same speaker appears in train and test. This *inflates* all reported figures, ours included. It is retained deliberately for comparability with

the base paper; a speaker-independent protocol would be stricter and would lower every number here.

- **Overfitting is unaddressed.** The 37-point train/validation gap indicates substantial headroom from regularisation alone.
- **The combined-leakage experiment was not completed**, so the compounded estimate (~88%) remains an extrapolation.

8.3 Future work

- Repeated-seed runs with significance testing.
- Regularisation study: the Dense head holds 66% of all parameters.
- Speaker-independent evaluation as a stricter secondary protocol.
- Investigating why AFW reduces confusion-pair errors more than CADL.

9. Efficiency study

Section 6.4 identified a 37-point train/validation gap. A follow-up study investigated whether regularisation could close it.

9.1 A correction to the overfitting diagnosis

The 37-point gap is measured at **epoch 14**. Training uses `EarlyStopping(restore_best_weights=True)` on validation loss, which restores **epoch 4**, where the gap is only 4.7 points. The *evaluated* model was therefore never badly overfit — early stopping was already performing most of the regularisation. This is why the interventions below produced a modest rather than a dramatic gain, and it corrects the interpretation offered in Section 6.4.

9.2 Validation sweep

Six configurations, novelties disabled, scored on **validation only**; the test set was not loaded during selection.

Configuration	Val accuracy	Params	Delta vs base
gap_reg_aug3	60.84%	2,540,167	+2.26
gap_reg	60.33%	2,540,167	+1.75
reg	59.40%	7,324,295	+0.82
gap	58.79%	2,540,167	+0.21
base	58.58%	7,324,295	+0.00
gap_reg_aug3_lr	58.58%	2,540,167	+0.00

The winner, `gap_reg_aug3`, combines a `GlobalAveragePooling` head with dropout 0.35/0.55, L2 1e-4 and 3x augmentation. Notably the pooling head **on its own** matched base accuracy (58.79% vs 58.58%) using 65% fewer parameters — the `Flatten(9,472)` to `Dense(512)` head carries 4.85 M parameters and contributes nothing measurable.

9.3 Test result

The winner was selected on validation and then evaluated on the test set **once**, alongside the base control.

Model	Accuracy	95% CI	Macro F1	MCC	AUC	Params
Base	57.95%	[55.99, 59.91]	0.5995	0.5059	0.8932	7,324,295
gap_reg_aug3	60.79%	[58.85, 62.73]	0.6271	0.5381	0.9063	2,540,167

+2.84 points with 65% fewer parameters. Every metric improves together — macro F1 0.5995 to 0.6271, MCC 0.5059 to 0.5381, AUC 0.8932 to 0.9063 — and the confusion-pair errors targeted by CADL fall from 188 to 137 (27.1% fewer) without CADL being enabled.

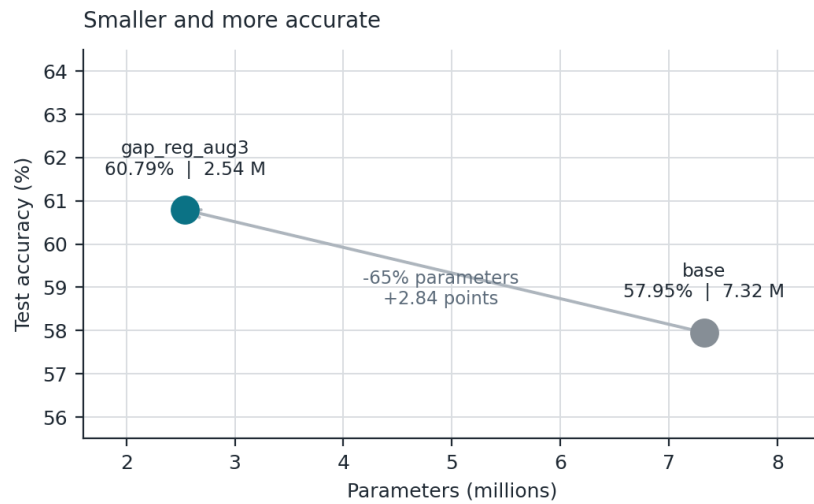


Figure 6 — The improved configuration is both smaller and more accurate.

For a project whose stated goal is a lightweight, edge-deployable model, a smaller *and* more accurate configuration is the more valuable of the two outcomes.

9.4 Per-corpus breakdown

Corpus	n	Base	gap_reg_aug3
TESS	553	96.75%	98.55%
RAVDESS	263	47.53%	55.13%
CREMA-D	1,513	47.12%	49.90%
SAVEE	104	35.58%	32.69%
Combined	2,433	57.95%	60.79%

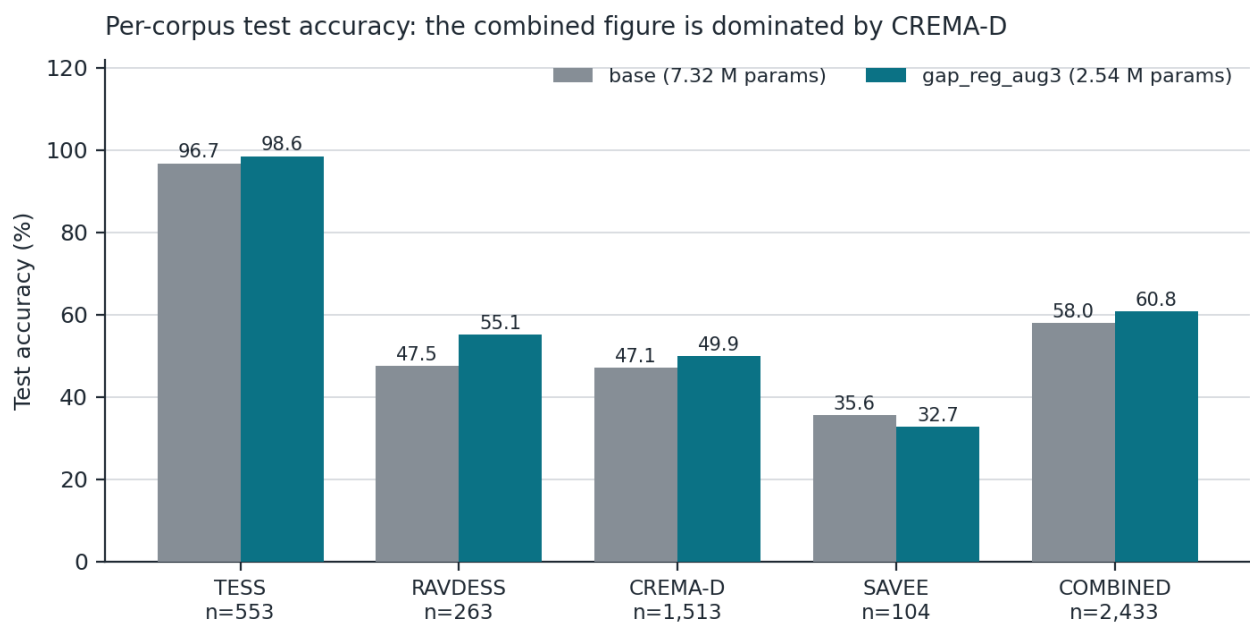


Figure 7 — Per-corpus test accuracy. The combined figure is dominated by CREMA-D, which is 62% of the test set.

This is the most informative table in the report. The same model scores **98.55% on TESS** and **49.90% on CREMA-D**. TESS is two speakers recorded in a studio with deliberately stereotyped delivery, and under the utterance-level split (Section 8.2) the same speakers appear in training. CREMA-D is 91 crowd-sourced actors with natural delivery and constitutes 62% of the test set.

Two consequences follow. First, the combined figure is essentially a CREMA-D figure. Second, and bearing directly on Section 7, a single-corpus result on TESS approaches 99% for a model managing barely 50% on CREMA-D. Published SER accuracies in the mid-90s are therefore explicable not only by the leakage mechanisms of Section 7 but also by **corpus composition** — evaluating on an easy, speaker-dependent corpus and reporting it as a general result.

SAVEE is the weakest at 32.69%, but with only 104 test samples that estimate is noisy, and it is the one corpus where the improved model does not beat base.

10. Conclusion

We implemented and ablated four lightweight novelties on a fused four-corpus SER dataset, delivering the component-wise analysis the base paper listed as future work. Three of four novelties improve on the baseline directionally, though none significantly at this sample size; CADL reduces its targeted confusion pairs by 8.5% as designed.

The project's principal contribution is methodological. The base paper's 94.91% could not be reproduced under a verified protocol, and we identify two concrete, independently measured leakage mechanisms — duplicated corpus mirrors and augment-before-split ordering — that inflate accuracy on this data by 14.7 and 24.3 points respectively. Both contamination rates were predicted before measurement and confirmed to within 0.2 points. We therefore report 57.95% as an honest baseline for this corpus combination, exceeding the closest comparable published result by 7.4 points.

The efficiency study adds a third result: 60.79% at 65% of the parameter count, which serves the lightweight-deployment goal better than the accuracy gain alone. The per-corpus breakdown supplies the sharpest single observation in this work — one model, 98.6% on TESS and 49.9% on CREMA-D — and makes clear that any SER accuracy quoted without its corpus composition is close to uninterpretable.

Appendix A — Artefacts and reproduction

```
results/ablation/runs/{base,afw,eaaa,mstc,cadl,full}/
  test_metrics.json      full metric suite
  test_classification_report.csv
  test_confusion_matrix{.csv,.png}
  training_curves.png, training_log.csv
  afw_weights_per_emotion.csv (AFW runs)
results/leak_dup/  duplicate-mirror experiment
results/leak_aug/  augment-before-split experiment
```

Reproduction:

```
py -3.10 -m venv .venv
.venv/Scripts/python.exe -m pip install -r requirements-dev.txt
.venv/Scripts/python.exe -m pytest tests/ -v
```

Kaggle notebooks, in order: 01_features (CPU, feature extraction), 02_train (GPU, six configurations), 03_leakage_test, 04_augment_before_split, 05_both_leaks.

Code, tests, notebooks and all artefacts: [**github.com/Eldorado5002/ser**](https://github.com/Eldorado5002/ser)